

Questionnaire: Resource Planning for Defect Management

Production: IBM Open XL C/C++ on Power (based on LLVM infrastructure)

Stakeholder: Siyuan Zhang, Project Manager

Part 1: Quantifying Workload & Effort

The goal here is to establish a baseline for converting the number of defects into a quantifiable workload.

1. Unit of Effort: When estimating the work required to fix a defect, what unit does your team typically use?

- ☐ Person-Hours
- ☒ Person-Days
- ☐ Person-Weeks
- ☐ Person-Months
- ☐ Other (please specify): _____

2. Average Effort per Defect: I know every defect is different, but for the purpose of creating a predictive model, could you provide an estimate for the average effort required?

- Approx. effort for a **Medium-complexity** defect (e.g., requires understanding module interactions, moderate debugging): 3 (please specify in the unit from Q1)
-

Part 2: Defining Risk Thresholds for Defect Inflow

This section will help me define the "Green-Yellow-Red" indicators. The goal is to understand at what point the number of new defects becomes a concern.

3. Green Zone (Normal Operations): In a typical week, what range of new incoming defects would you consider to be normal and manageable for the team?

- Approx. less than 10 new defects per week.

4. Yellow Zone (Caution / Requires Attention): At what number of new defects in a single week would you start to become concerned and feel the need to investigate or monitor the situation more closely?

- Approx. 10 new defects per week.

5. Red Zone (Alert / Action Required): At what number of new defects in a single week would you consider it a critical situation, potentially requiring you to re-allocate resources from other tasks or adjust project timelines?

- Approx. more than 10 new defects per week.

Note: among new defects, we care more about regression defects' number. If a new defect is of new functionality, it will still bring attention but it can also fall into the scope of the new functionality's development phase. Regression defect brings more attentions since it means the merged code (no matter from new functionality or defect fixing) causes quality problem.

Part 3: Data & Process Context

6. Defect Tracking System: What are the primary sources of defects on the Open XL Compiler project?

- ☒ Customer reports (e.g., via support channels)
- ☒ Upstream LLVM community (e.g., GitHub Issues)
- ☐ Private GitHub repository
- ☐ Bugzilla
- ☐ Other (please specify): _____

7. Key Data Fields: When exporting defect data, which of the following fields are typically required?
(Please check all that apply)

- ☒ Defect ID
 - ☒ Creation Date (or Open Date)
 - ☒ Status (New, Assigned, Closed)
 - ☒ Priority (Highest, High, Medium, Low)
 - ☒ Component (e.g., clang:frontend, llvm:codegen, mlir, lldb)
 - ☐ Backend (e.g., PowerPC, x86, RISC-V)
 - ☐ Other (please specify): __severity, version found in, reported by (customer, or QA, or Dev)_____
-